Niranjan Kamalakannan

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EDUCATION

University of Maryland, Robert H. Smith School of Business **College Park, MD, USA**
Master of Quantitative Finance, GPA 3.8/4 May 2026

- Awarded the prestigious "Terrapin Scholar"
- Relevant coursework: Financial Data Analytics, Financial Management, Advanced Capital Markets, Machine Learning in Finance, Derivative Securities, Valuation, Financial Mathematics, Econometrics Modeling, and Fixed Income Derivatives
- Leadership: Student Ambassador for MQF, Vice President of Operations in Smith Masters Finance Association
- Teaching Assistant for Machine Learning and Financial Data Analytics courses

PSG College of Technology **Coimbatore, TN, India**
Bachelor of Engineering in Mechanical Engineering, **GPA 7.67/10 (First Class)** April 2023

- Alumni Coordinator, 2023 class; Leadership roles: Co-founder of Finverse (Finance Club)

SKILLS AND CERTIFICATIONS

- **Programming & Databases:** Python (pandas, NumPy, scikit-learn, statsmodels, Vectorbt), SQL, R, Excel, WRDS, Bloomberg, LSEG
- **Analytics & Modeling:** Derivatives pricing (Black-Scholes, Binomial, Monte Carlo, Heston)(GitHub), Time series modeling (GitHub)
- **Machine Learning & Statistics:** Regression methods, feature engineering, hypothesis testing, probability distributions
- **Data Visualization:** Python (Matplotlib, Seaborn, Plotly), Tableau
- **Certifications:** CFA Program – Level 1, Bloomberg Market Concepts (BMC)

WORK EXPERIENCE

Global Equity Fund **College Park, Maryland**
Equity Research Analyst May 2025 –Present

- Analyzed and valued industrials sector equities for Global Equity Fund managing $750K+ across 30+ positions.
- Developed and presented investment theses on industrials sector equities including Global Ship Lease, analyzing shipping market fundamentals, financial metrics, and competitive dynamics to support portfolio decisions.

Google-Sponsored Experiential Learning Program **College Park, Maryland**
Risk Analyst January 2025 – March 2025

- Implemented an AI-based credit risk framework integrating **NLP and LLMs**; analyzed over 60 banks and fintech companies across six quarters using SEC filings and earnings call transcripts.
- Engineered credit risk scoring models using Google Gemini and Python, achieving 39% prompt accuracy across multiple trials and identifying distressed banks in a 2008 crisis backtest with high fidelity.
- Quantified credit risk trends via sentiment analysis across 16 weighted categories, generating 1–5 risk scores per institution.

PROJECT EXPERIENCE

Stochastic Interest Rate Modeling & Bond Pricing (Python) (GitHub)

- Constructed initial term structures for fixed-income modeling by calibrating Nelson–Siegel, Nelson–Siegel–Svensson, and Cubic Spline models to 1M–30Y yield data, achieving an RMSE of 4 bps (NSS) vs 5 bps (NS).
- Developed Vasicek and CIR stochastic interest-rate models to capture realistic mean-reverting dynamics required for bond pricing and simulation, improving the realism of rate-path generation.
- Applied Maximum Likelihood Estimation (MLE) to calibrate mean reversion, long-term mean, and volatility parameters, followed by model comparison and scenario analyses to assess regime-dependent performance in fixed-income portfolios.

Credit Risk Modeling (PD, LGD, EAD) – Basel III (Python) (GitHub)

- Built an end-to-end credit risk modeling pipeline using 900K LendingClub loans to estimate PD, LGD, EAD, and Expected Loss; ensured alignment with Basel III IRB regulatory standards for capital adequacy and risk governance.
- Engineered predictive features using Weight of Evidence (WoE) and Information Value (IV); trained logistic and linear models for PD (AUROC-72%, Gini-0.44, KS-0.3208), LGD (AUROC-64.8%, RMSE-9%), and EAD (53% correlation with actual CCF).
- Developed a credit scorecard by transforming logistic regression outputs into scaled scores; optimized risk cutoffs using TPR/FPR/KS-statistic and built PSI-based monitoring to ensure long-term model stability and drift detection.
- Estimated total portfolio-level Expected Loss of $426.9M over $6.66B in funded exposure (6.41% EL rate); validated all models using out-of-sample data and benchmarked the entire modeling pipeline with Basel III IRB regulatory benchmarks.

Market Risk: Portfolio VaR and CVaR Modeling and Backtesting (Python, Excel) (GitHub)

- Quantified tail-risk for a $1M 5-stock portfolio by building VaR/CVaR models using Historical, Variance-Covariance, and Monte Carlo methods, producing 1-day risk estimates at 99%, 95%, and 90% confidence levels.
- Validated portfolio risk via Basel Traffic Light (3 breaches, Green Zone) and Kupiec's POF test, improving Monte Carlo simulations using Cholesky decomposition to model correlated return shocks and cross-asset dependencies.
- Evaluated portfolio resilience under adverse markets by running sensitivity and stress tests for 1%, 5%, and 10% VaR shocks, identifying potential drawdowns and enhancing market-risk oversight.